

Computer Security

The protection given to an automated information system to attain the applicable objectives of preserving the integrity, availability & confidentiality of information system resources.

Confidentiality $\Rightarrow$ Protection of data from unauthorized disclosure.

Integrity $\Rightarrow$ The assurance that data received are exactly as sent by an authorized entity without any modifications, insertions or deletion.

Availability $\Rightarrow$ Ensures that systems are responsive & that service is not withheld from authorized user.

Levels of Impact

Low Impact

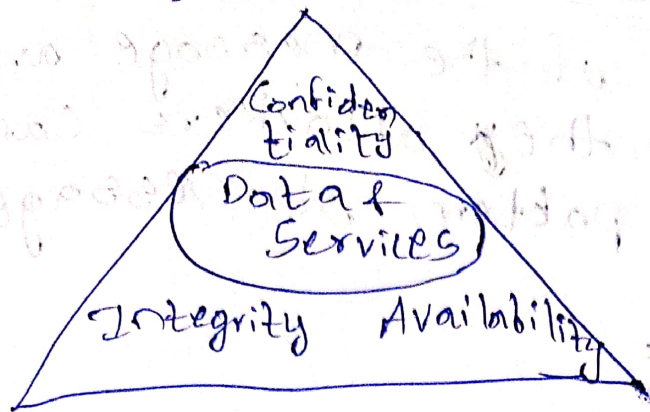
* The loss can be anticipated to have a minimally negative impact on an organization's operations, assets, or people

Moderate

* Will have a serious negative impact

High

* Will have a terrible negative impact



Three Aspects of Security

* Open Systems Interconnection (OSI) security architecture offers a systematic framework.

- 1) Security Attacks $\Rightarrow$ Any action that compromises the security of info owned by an organization.
- 2) Security mechanisms $\Rightarrow$ A process that is designed to detect, prevent, or recover from a security attack.
- 3) Security services $\Rightarrow$ enhances the security of the data processing systems.

Security attacks

- 1) Passive
- 2) Active

1) Passive Attacks

* The contents of the message are released. Also, the attackers can observe the pattern of messages.

2) Active Attacks

* Masquerade

→ The messages can be edited and sent to the dest.

* Replay

→ Captures the messages from src to dest, later replays the message to dest.

* Modification of messages

* Denial of Service

→ The services of server are disrupted by the attacker.

Security Services

1) Authentication

2) Access Control → prevention of unauthorized use of a resource.

3) Data Confidentiality

4) Data Integrity

5) Non Repudiation

* Non repudiation, origin
→ Proof that message was sent by the specified party.

* Non repudiation, Destination
→ Proof that the message was received by the specified party.

Security Mechanisms

1) specific security mechanisms

* Incorporated into the appropriate protocol layer to provide some of the OSI security.

→ Encipherment:- The use of mathematical algorithms to transform data into a form that is not readily intelligible.

→ Digital Signature:- A cryptographic transformation of the data unit to prove the integrity of the data.

→ Access Control

→ Data Integrity

→ Authentication Exchange ⇒ A mechanism intended to ensure the identity of an entity by means of information exchange.

→ Traffic Padding ⇒ The inserts of bits into gaps in a data stream to frustrate traffic analysis attempts.

→ Fonting Control

→ Notarization: The use of a trusted third party to assure certain properties of a data exchange.

2) Pervasive Security Mechanisms

* These are not specific to any particular OS security service or protocol layer.

→ Trusted Functionality

→ Security Label

→ Event Detection

→ Security Audit Trail

→ Security Recovery

Antivirus

* It is a security program that detects, blocks, quarantines, and malicious software before it damages a system.

* Acts as the first line of defense.

Antivirus detection Techniques

1) Signature-based Detection

* Identifies malicious software by comparing files against a database of known malware signatures or unique digital fingerprints.

2) Heuristic Analysis

* Identifies previously unknown malware ~~as~~ by analyzing suspicious code patterns, programming structures, & behaviors.

→ Static M.A:- Examines the program without running it.

→ Dynamic M.A:- Runs the program in a controlled environment.

3) Behavior-based detection

* Monitors the real-time activities of a running application.

4) ML-based detection

* This uses AI models trained on millions of benevolent & malicious files to predict whether a file is malware.

5) Sandboxing

* Executes suspicious files inside an isolated, virtual environment where their behavior can be safely observed.

Antivirus Bypassing

* Refers to techniques used by attackers to avoid detection by security software.

1) Code obfuscation:-

Process of making a program's Code difficult to understand.

2) Packing

Process of compressing or encoding an executable file.

3) Encryption

Converts readable program code into an unreadable form.

4) Polymorphism

* This is a technique which malware changes its code or appearance every time it copies or spreads.

5) Fileless malware

* Performs malicious actions without creating traditional executable files.

6) Delayed execution

* Here malware waits for a period or a specific event.

Limitations of Antivirus

- Zero-Day Threats
- Sophisticated Fileless Attacks
- Encrypted Payloads.
- False Positives.

Prevention Techniques

- Endpoint Detection & Response (EDR)
- Behavior Monitoring
- Application Whitelisting ⇒ Allows only approved and trusted software to run on a computer.
- timely updates.
- Threat Intelligence
- User Awareness

Web Browser Exploitation

- * Attacker targets browsers, plugins, extensions, cookies, sessions.

Browser Exploitation Techniques

* 1) Cross-site Scripting (XSS)

- * This is a web security vulnerability that allows an attacker to inject malicious scripts into a trusted website.

2) Cross-site Request Forgery (CSRF)

- * This tricks a logged-in user into performing an unwanted action on a trusted website.

3) Drive by Downloads

- * A malicious download that occurs automatically when a user visits a malicious website.

4) Malicious browser extensions

* This is an add-on that appears useful but secretly performs unauthorized activities such as stealing data or monitoring user activity.

5) Clickjacking

* This is an attack where a user is tricked into clicking a hidden or disguised web page element without realizing it.

6) Session hijacking

* Attacker steals or takes over a user's active login session to gain unauthorized access to an account.

Browser Security Mechanisms

1) Same Origin Policy (SOP)

* Restricts a web page from accessing data belonging to another website.

2) Content Security Policy (CSP)

* Browser security feature that controls which resources a webpage is allowed to load & execute.

3) HTTPS & HSTS (HyperText Transfer Protocol Secure)

* HTTPS is the secure version of HTTP that encrypts communication b/w the browser & the web server using SSL/TLS.

4) Secure Cookies

* A cookie is a small piece of data stored by the browser to remember information such as login sessions & user preferences.

Classical Encryption Techniques

Substitution Techniques

* Replacing plain text letters with some other letter, number or symbol is called substitution.

Types

1) Monoalphabetic

2) Polyalphabetic

Casas Cipher

* Each letter in the alphabet is added with the given key.

Encryption :- $C = (P + K) \bmod 26$

Decryption :- $P = (C - K) \bmod 26$

P = Plaintext; C = Ciphertext; K = Key.

Ex: - CAR, key = 3
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$$C = (2+3) \% 26 \quad | \quad A = (6+3) \% 26 \\ = 5 \% 26 = 5 = F \quad | \quad = 3 \% 26 = 3 = D$$

$$R = (17+3) \% 26$$

$$= 20 \% 26$$

$$= 20 = U$$

Playfair cipher

* Construct a 5x5 matrix

* Fill the matrix with the alphabets in the text, if the letters are repeating in the key then write it only once.

* Fill the rest of the matrix with the alphabets that are not present in the key.

* Break the message into pairs of letters.

* If a pair has repeated letters, insert a filler letter (usually X).

* If a message has odd no. of letters add X at the end.

Ex:- Plaintext = BALLOON
Key = MONARCHY

M	O	N	A	R	BA LX LO ON
C	M	Y	B	D	IB SU MP NA
E	F	G	I	J	
L	P	Q	S	T	
U	V	W	X	Z	

Ciphertext = IB SUPMNA

Rules

- Same Row ⇒ Replace each letter with the one to its LEFT
- Same Column ⇒ Replace each letter with the one ABOVE it or below.
- Rectangle rule ⇒ Replace each letter with the letter in the same row, but in the column of the other letter.

Ex:- Plaintext = HACKER
Key = REACH

B	R	E	A	C
M	D	F	G	I
K	L	M	N	O
P	Q	R	S	T
V	W	X	Y	Z

H	A	C	K	E	R
A	D	D	O	A	E

P:- FIREWALL

K:- SECURE

~~S E C U R~~
~~X A B D F~~
~~G H I J K L~~
~~M N O P Q~~
~~T V W X Z~~

FI / RE / WA / LX
BL / UC / VB / GA

S E C U R
A B D F G
H I J K L M
N O P Q T
V W X Y Z

FI / RE / WA / LX
BL / SC / VB / KX

BL SC VB KY

EX:-

T R A N S

F I E B C
D G H I J K
L M O P Q
U V W X Y Z

T R A N S

F I E B C D
G H I J K L
M O P Q U
V W X Y Z

UPLOAD
MOUSE

K E X B O
A R D C F
G H I J L M
N P Q S T
U V W X Z

CI / PK / ER / TEXT
DL / NR / PO / ZS
RM

DL VP RM PO ZS



HILL Cipher

* Each letter in the plaintext is arranged in matrix form and is multiplied with key matrix is encryption to get ciphertext.

$$\text{Encryption: } C = (P * K) \% 26$$

$$P = (C * K)^{-1} \% 26$$

P:- HELP

$$K = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix}$$

$$H = 7$$

$$E = 4$$

$$L = 11$$

$$P = 15$$

$$\begin{pmatrix} H \\ E \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 7 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \times 7 + 3 \times 4 \\ 2 \times 7 + 5 \times 4 \end{pmatrix} = \begin{pmatrix} 33 \\ 34 \end{pmatrix} \% 26$$

$$\begin{pmatrix} L \\ P \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 11 \\ 15 \end{pmatrix} = \begin{pmatrix} 33 + 45 \\ 22 + 75 \end{pmatrix} = \begin{pmatrix} 78 \\ 97 \end{pmatrix} \% 26 = \begin{pmatrix} 0 \\ 19 \end{pmatrix} = \begin{pmatrix} 4 \\ 19 \end{pmatrix}$$

Ciphertext = HIAT

Encryption

$$\begin{aligned} \Delta \text{Key} &= ad - bc \\ &= 3 \times 5 - 3 \times 2 \\ &= 15 - 6 \\ &= 9 \% 26 = 9 \end{aligned}$$

$$\text{Now } (9 \times 9) \% 26 = 1$$

$$\boxed{x = 3}$$

$$\text{Adj. Key} = \begin{bmatrix} 5 & 3 \\ -2 & 3 \end{bmatrix} \% 26 = \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix}$$

$$3 \times \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix} = \begin{bmatrix} 15 & 69 \\ 72 & 9 \end{bmatrix} \div 26$$

$$= \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

$$\begin{pmatrix} M \\ I \end{pmatrix} = \begin{pmatrix} 7 \\ 8 \end{pmatrix} \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \div 26 = \begin{pmatrix} 105 + 136 \\ 140 + 72 \end{pmatrix} \div 26$$

$$= 17 = M$$

$$= 14 = E$$

$$\begin{pmatrix} A \\ T \end{pmatrix} = \begin{pmatrix} L \\ P \end{pmatrix}$$

Decrypted Text = HELD

Vigenere Cipher

1) Choose a keyword

2) Repeat the keyword

3) Convert letters to numbers

4) Encrypt $C_i = (P_i + K_i) \div 26$

5) Decrypt $P_i = (C_i - K_i) \div 26$

Ex: $P = \text{Examination}$
 $K = \text{Hacker}$

E	X	A	M	I	N	A	T	I	O	N
4	23	0	12	8	13	0	19	8	14	13

H	A	C	K	E	R	H	A	C	K	E
7	0	2	10	4	17	7	0	2	10	4

11	23	2	22	12	4	7	19	10	24	17
L	X	C	W	M	E	H	T	K	Y	R

Ciphertext $\Rightarrow$ L X C W M E H T K Y R

Ex:- P = BACKUP

B	A	C	K	U	P
2	0	2	10	20	15
R	E	S	T	O	R
17	4	18	19	14	17

18	4	20	3	8	16
S	E	U	D	I	Q

A = 0

B = 1

C = 2

D = 3

E = 4

F = 5

G = 6

H = 7

I = 8

J = 9

K = 10

L = 11

M = 12

N = 13

O = 14

P = 15

Q = 16

R = 17

S = 18

T = 19

U = 20

V = 21

W = 22

X = 23

Y = 24

Z = 25

One-Time Pad

* Same as Vigenere cipher, but the length of Plaintext is same as length of key.

Ex:- P = HELLO
K = XMKL

H	E	L	L	O
7	4	11	11	14

X	M	L	K	L
23	12	2	10	11

4	16	13	21	25
---	----	----	----	----

E	Q	N	V	Z
---	---	---	---	---

$\Rightarrow$ Cipher

Vernam Cipher

→ Plaintext letters are XORed with key letters to get cipher text

Encryption $\Rightarrow C_i = P_i \oplus K_i$

Decryption $\Rightarrow P_i = C_i \oplus K_i$

Ex:- P:- OAK
K:- SON

O	A	K
79	65	75
01001111	01100001	01001011
S	O	N
83	79	78
01010011	01001111	01001110
<u>00011100</u>	<u>00001110</u>	<u>00000101</u>
		↙
		Cipher Text:-

Transposition Techniques

* Rearranging letters of plain text is called as transpositions.

1) Rail fence

P:- HELLO TIM

K:- 2 $\Rightarrow$ 2 rails

R1: H L O I

R2: E L T M

Ciphertext $\Rightarrow$ H L O I E L T M

Ex:- P = DEFEND THE WALL
K = 3

D			N			E			L
E		E		D	H	W		L	
F				T			A		

Ciphertext $\Rightarrow$ D N E L E E D H W L F T A

Columnar Transposition

1) Single Columnar Transposition

Ex:- P:- attack at four

K:- 2 3 4 1

2 3 4 1

a t t a

c k a t

f o u r

$\Rightarrow$ arrange according to the key column positions

a t r a c f t k o t a u

$\Rightarrow$ ciphertext.

Ex:- ATTACK AT FOUR $\Rightarrow$ P

Key: ZEBRA

5 3 2 4 1 $\Rightarrow$ Take alphabetically.

A T T A C

K A T F O

U R X X X

Cipher $\Rightarrow$ C O X T T X T A R A F X A K U

2) Double Columnar Transposition

P:- ATTACK AT FOUR

K1:- 4 3 1 2 5 6 7

K2:- 3 1 4 2 7 5 6

4 3 1 2 5 6 7

A T T A C K A T

T F O U R X X

Middle Text $\Rightarrow$ T O A U T F A T C R K X A X

3 1 4 2 7 5 6

T O A U T F A

T C R K X A X

Cipher $\Rightarrow$ O C U K T T A R F A A X T X